



ENERVISION

Empowering Smart Energy Insights



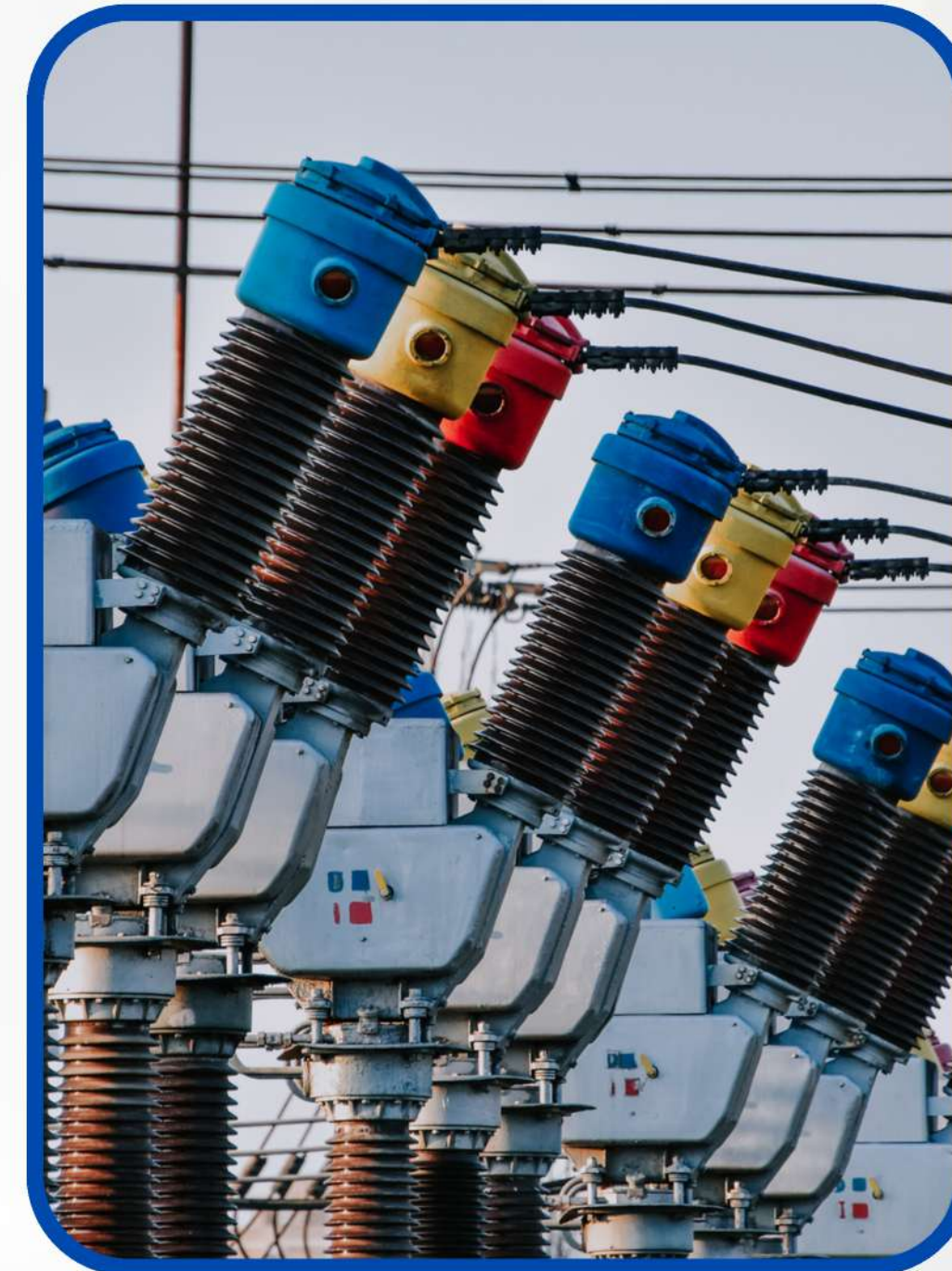
EnerVision

ABOUT ENERVISION

EnerVision blends IoT sensors and **95%-accurate AI to transform energy monitoring and optimization.** With energy costs rising and India's demand growing **4.2% annually**, we deliver continuous insights far beyond traditional audits.

Organizations **achieve 20–25% Energy savings** and up to **25% carbon reduction.** Our proven prototype and strong multi-sector validation position us to lead **India's ₹10,000 crore market.**

Backed by deep expertise in IoT, AI, and energy engineering, EnerVision turns energy use into strategic advantage.



EnerVision goes beyond **monitoring by using IoT sensors and AI to analyze real-time data** every 6 seconds and deliver prescriptive, context-aware recommendations.

Our system **identifies issues like HVAC zones consuming 22% extra power**, traces root causes using weather, occupancy, and equipment history, and suggests the most cost-effective fix.

We also integrate sustainability **metrics—carbon reduction, ESG reporting, and renewable planning—bringing operational efficiency** and environmental impact into one platform.

WHAT WE OFFER **ALWAYS**

1



2



3



Features

AI-DRIVEN PREDICTIVE ANALYTICS

"Smart insights
today for a
sustainable, cost-
efficient tomorrow."

REAL-TIME ENERGY MONITORING

"See your energy,
save your costs—live,
anytime, anywhere."

SUSTAINABILITY AND CARBON FOOTPRINT REPORTING

"Track, reduce, and
thrive with eco-friendly
energy practices."

CUSTOMIZED ENERGY EFFICIENCY SOLUTIONS

"Tailored strategies to
power efficiency and
cut waste."

ENERVISON CORE EFFICIENCY SERVICES

- Provides deep real-time visibility into power patterns, peak loads, and appliance behavior, enabling quick detection of abnormalities.
- Uses AI-driven forecasting to predict consumption trends, optimize usage, and reduce operational energy costs.

1

REAL TIME ENERGY
MONITORING

2

AI POWERED
PREDICTIVE ANALYSIS

3

APPLIANCE LEVEL
CONSUMPTION
ANALYSIS

ENERGY EFFICIENCY
RECOMMENDATION

1

CARBON FOOTPRINT
AND CARBON CREDIT
REPORTING

2

ARCHITECTURAL
DESIGN ANALYSIS FOR
ENERGY OPTIMIZATION

3

ENERVISION ENERGY DRIVEN SERVICES

- Provides precise energy-efficiency recommendations, identifies major loss points, and calculates real-time carbon footprint with carbon-credit readiness for sustainability reporting.
- Integrates architectural design analysis to optimize building layout, airflow, equipment positioning, and sensor placement—reducing power losses and improving long-term energy performance.

WE PROVIDE TO

MARKETPLACE

CUSTOMER SEGEMENT

2M

India's 1.5M schools and 39k colleges spend ₹50L–₹2Cr annually on energy, making them a high-impact segment for savings.

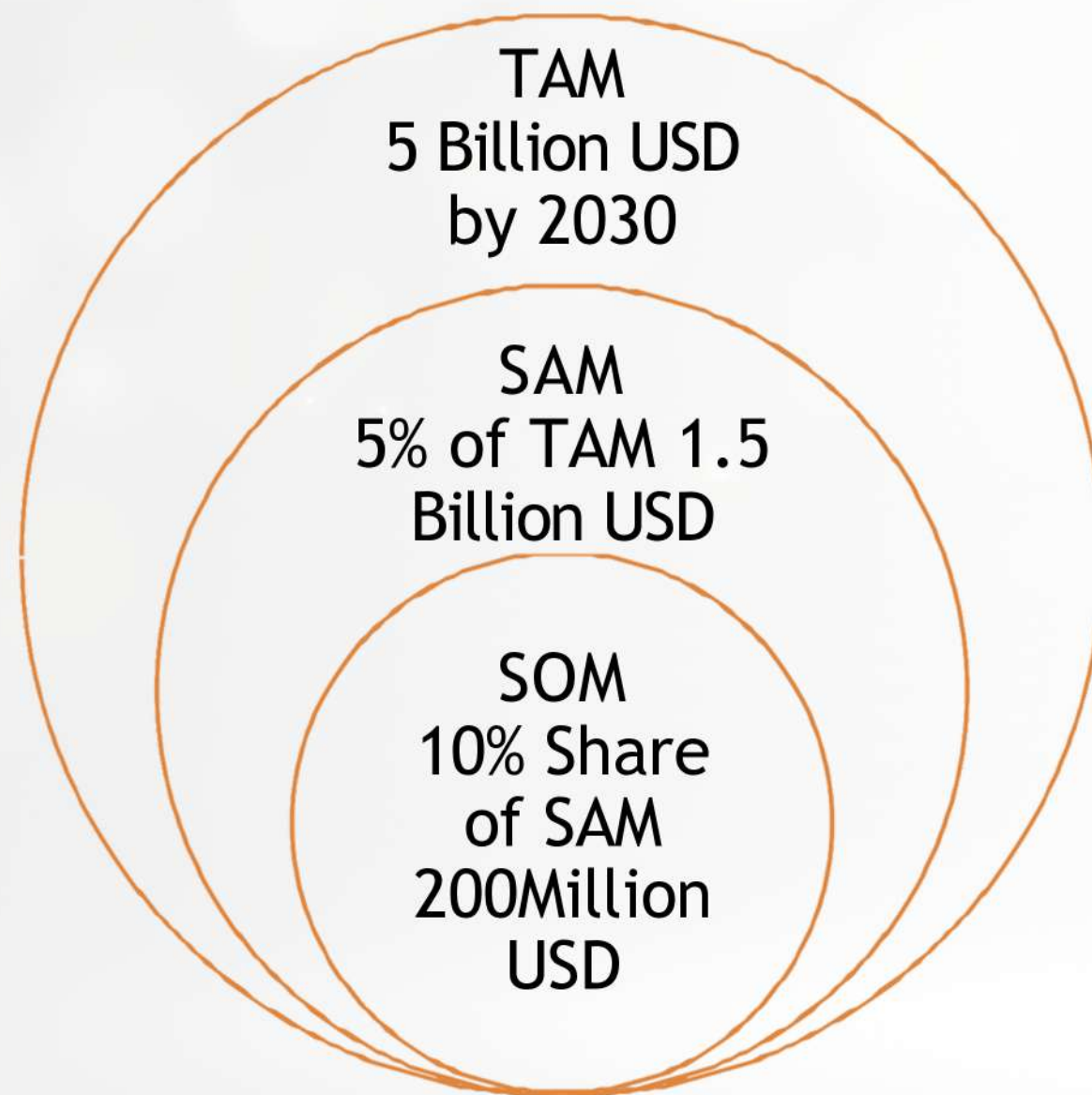
600M
Sq.ft

Commercial office spaces consume massive power for 24/7 operations, representing a fast-growing ₹5,000 crore market.

43%

Industries like textiles, pharma, and automotive consume 40% of India's industrial energy—where even 2–3% savings shift profitability.

MARKET SIZE



Total Addressable Market (TAM):

\$5 billion (~₹40,000 crores) by 2030.

Represents the entire global market for energy auditing systems across industries like manufacturing, healthcare, education, retail, and commercial offices.

Serviceable Available Market (SAM):

\$1.2 billion (~₹10,000 crores) by 2030.

Focuses on the Indian energy management .

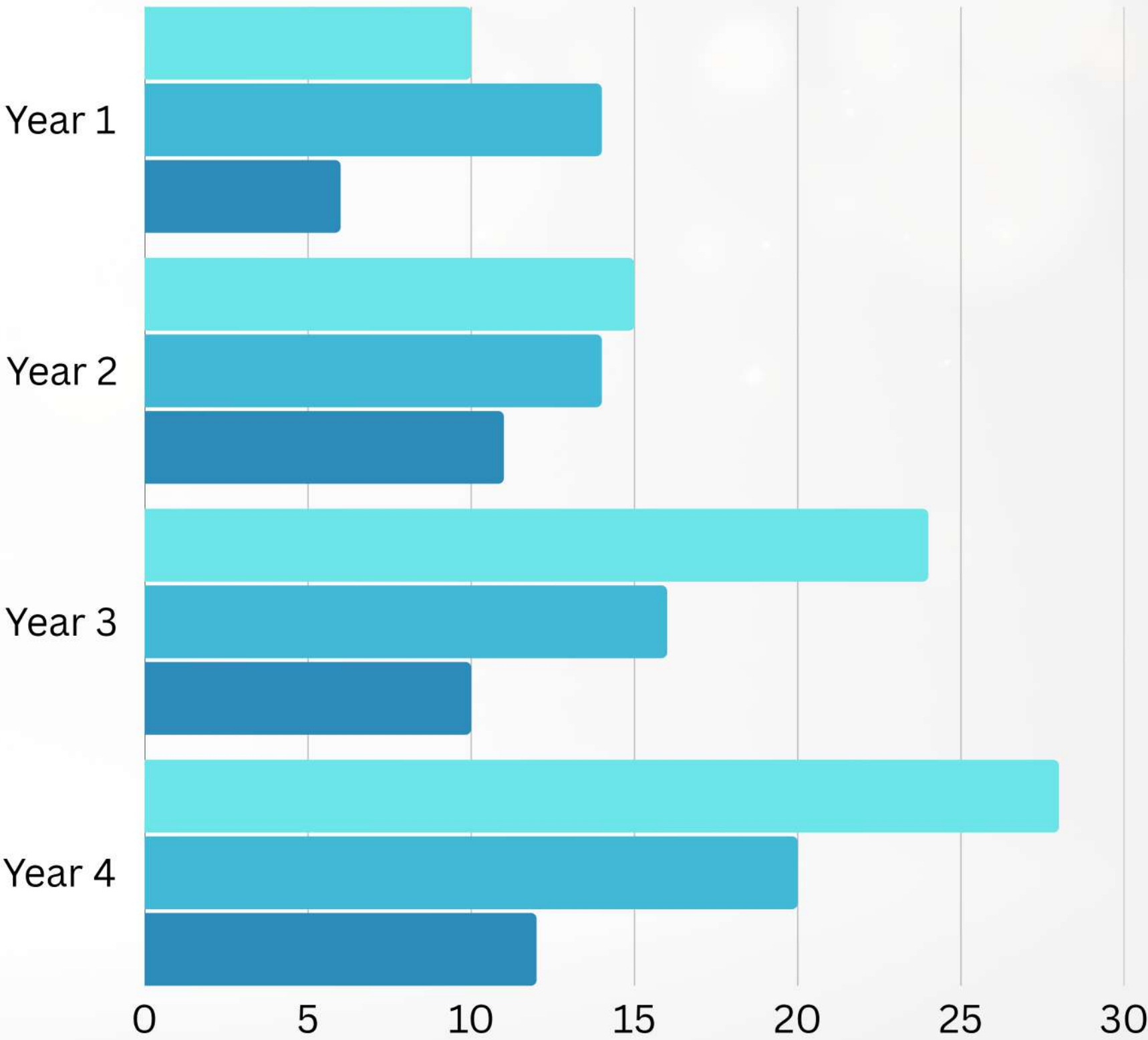
Serviceable Obtainable Market (SOM):

\$60 million (~₹500 crores).

Represents 5% of the SAM, achievable by Enervision within 2–3 years, accounting for competition, scalability, and operational capabilities.

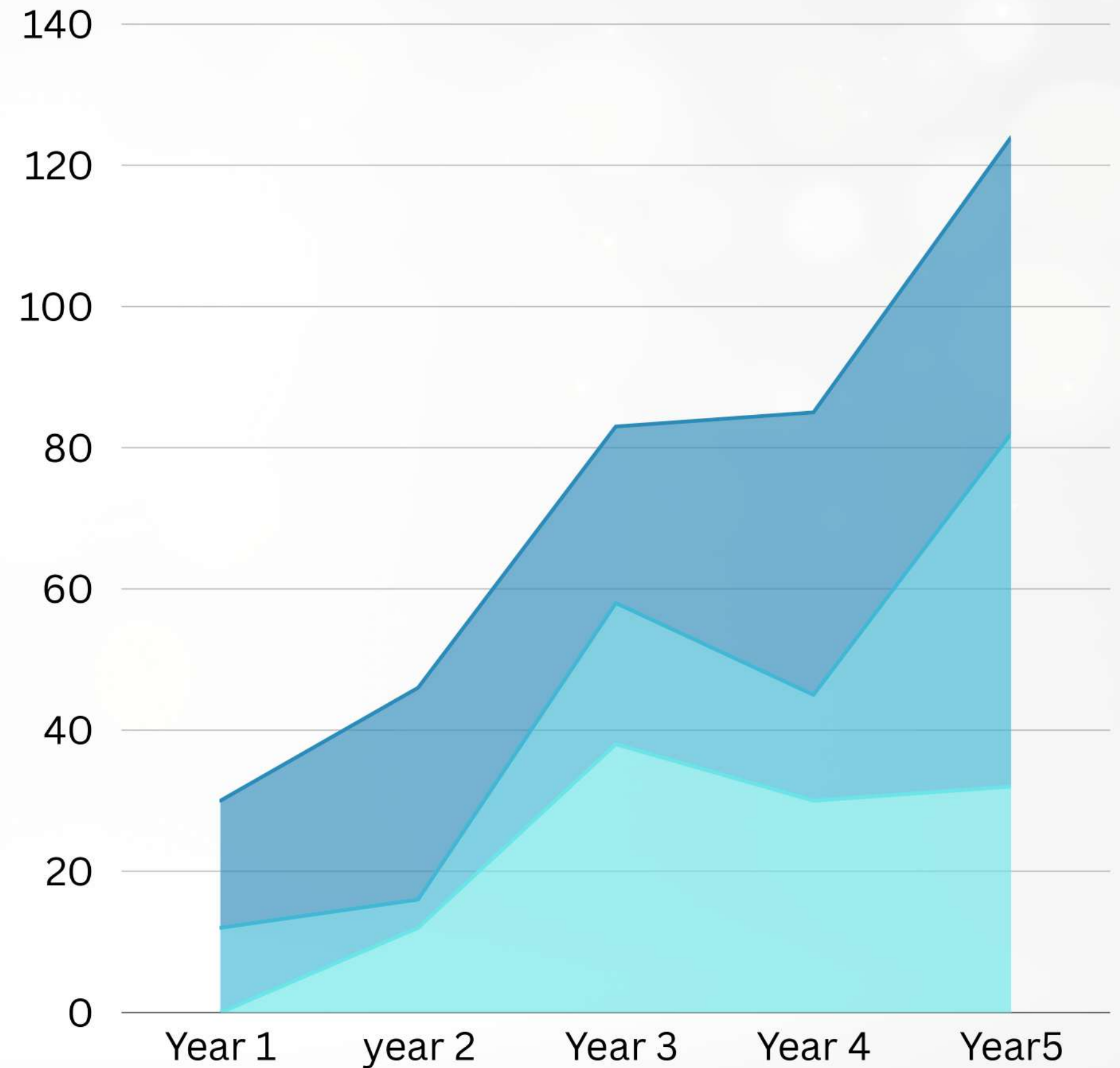
IMPACT FLOW CHART

- Every 1% reduction in annual energy usage results in **2–3% reduction in long-term operational cost**, due to compounding savings, reduced maintenance, and lower peak-time penalties.
- Buildings that adopt smart energy monitoring report **15–25% reduction in electricity consumption within the first year**, based on global facility-management benchmarks.
- Predictive maintenance driven by **energy analytics reduces equipment failure by 30–40%, increasing lifespan of HVAC systems** and industrial machinery by up to 5–7 years.
- Improved power factor and optimized load balancing can **save 8–12% in monthly bills**—especially in commercial and industrial environments.
- Carbon footprint reduction typically scales as **0.82 kg of CO₂ per kWh saved**, meaning a building saving just **50,000 kWh/year avoids ~41 tons of CO₂**, making it eligible for multiple carbon-credit schemes.
- Over 5 years, **cumulative savings can reach 60–70% of a building’s annual electricity budget**, once compounding, efficiency upgrades, and reduced downtime are factored in.

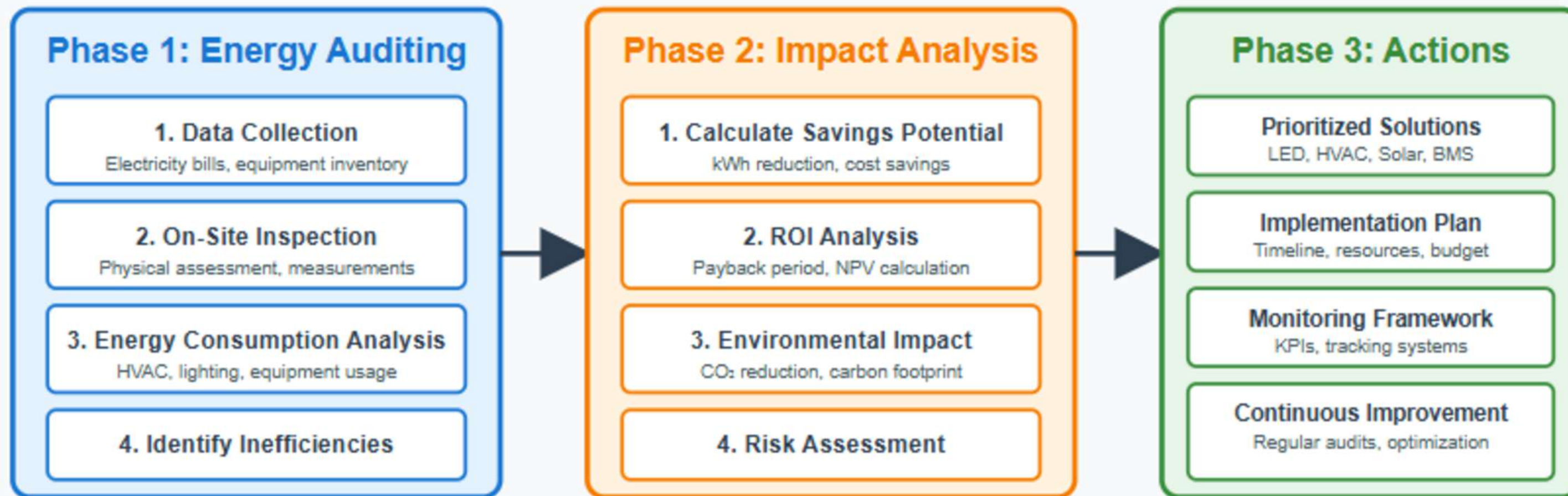


ENERGY SAVING

- Immediate reduction as the system identifies and fixes wastage in lighting, HVAC, and machinery within weeks.
- AI-driven optimization improves monthly, learning usage patterns to steadily increase savings.
- Predictive maintenance prevents energy-heavy equipment failures, reducing overconsumption long-term.
- Architectural & operational improvements refine sensor placement and building flow for deeper efficiency gains.
- Compounding savings accumulate year after year, cutting total energy costs by 50–70% over 5–10 years.



Energy Auditing and Impact Analysis Framework



Key Impact Metrics

Financial Impact

- Annual cost savings
- Investment required
 - Payback period
 - Net present value
- Internal rate of return

ROI: 2-5 years typical

Schools: ₹50L-₹2Cr savings

Energy Reduction

- kWh saved annually
- % reduction in consumption
- Peak demand reduction
- Load factor improvement
- Energy intensity metrics

Target: 20-40% reduction

Industrial: Small % = Big impact

Environmental

- CO₂ emissions reduced
- Carbon credits earned
- Green certifications
- Renewable energy %
- Sustainability rating

Industrial: 40% of total

IT Parks: LEED certification

Operational Benefits

- Equipment lifespan
- Maintenance cost reduction
- Comfort improvement
- Productivity gains
- Brand reputation

Intangible value

Offices: Employee satisfaction

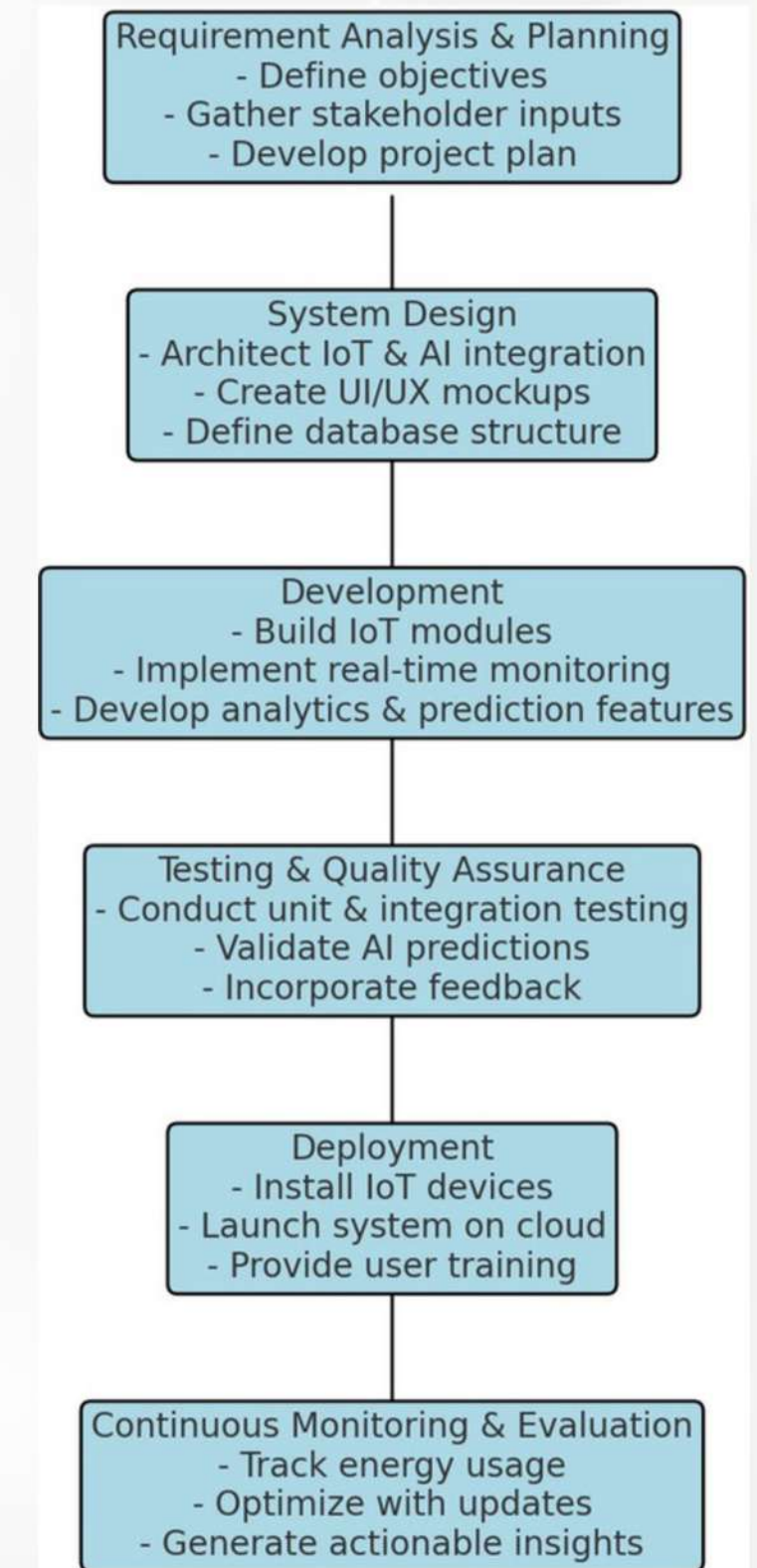
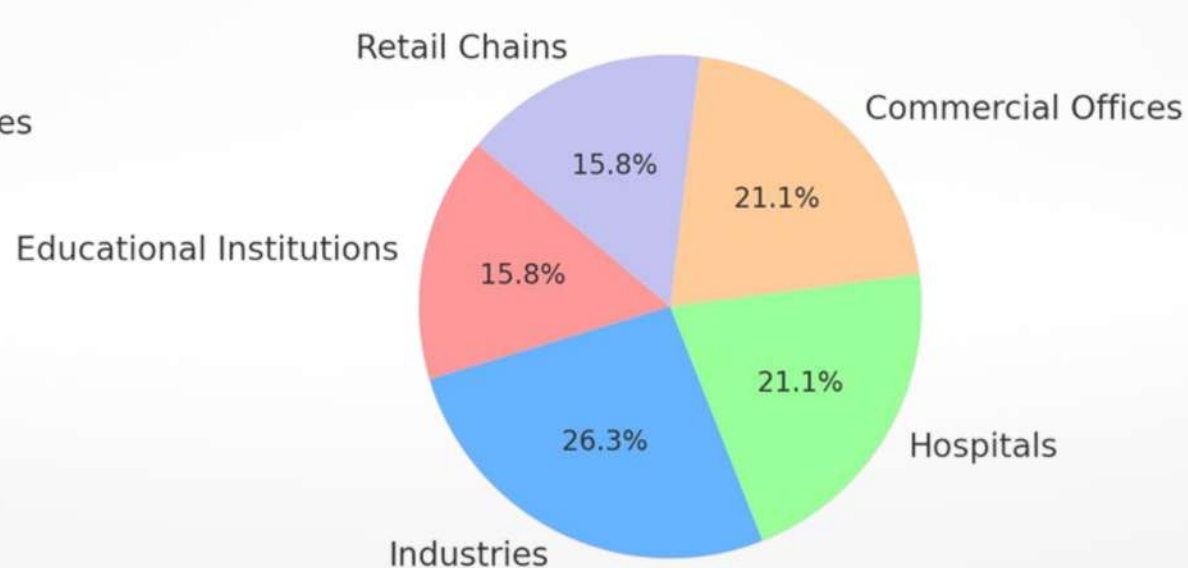
ENERVISION UNIQUENESS

- **Full-Stack Energy Intelligence + Architectural Design Analysis:** EnerVision goes beyond monitoring by combining **real-time IoT sensing, predictive AI, and architectural design analysis** to identify structural inefficiencies (lighting layout issues, HVAC zoning flaws, load distribution errors) and provide actionable redesign recommendations that boost building-level efficiency.
- **Maximum Savings With Predictive Optimization:** Our platform delivers **20–25% energy reduction** using continuous monitoring, instant anomaly detection, equipment-health prediction, and sector-specific optimization—doubling the impact of traditional audits while remaining easier and more customizable than global solutions like Google EMS.
- **Integrated Sustainability + Carbon Credit Generation:** EnerVision automatically **tracks carbon footprint**, measures emission reductions, and converts **verified savings into tradable carbon credits**, unlocking new revenue streams while ensuring ESG compliance, sustainability reporting, and alignment with global climate frameworks.

Energy Audit Efficiency Improvement by Sector

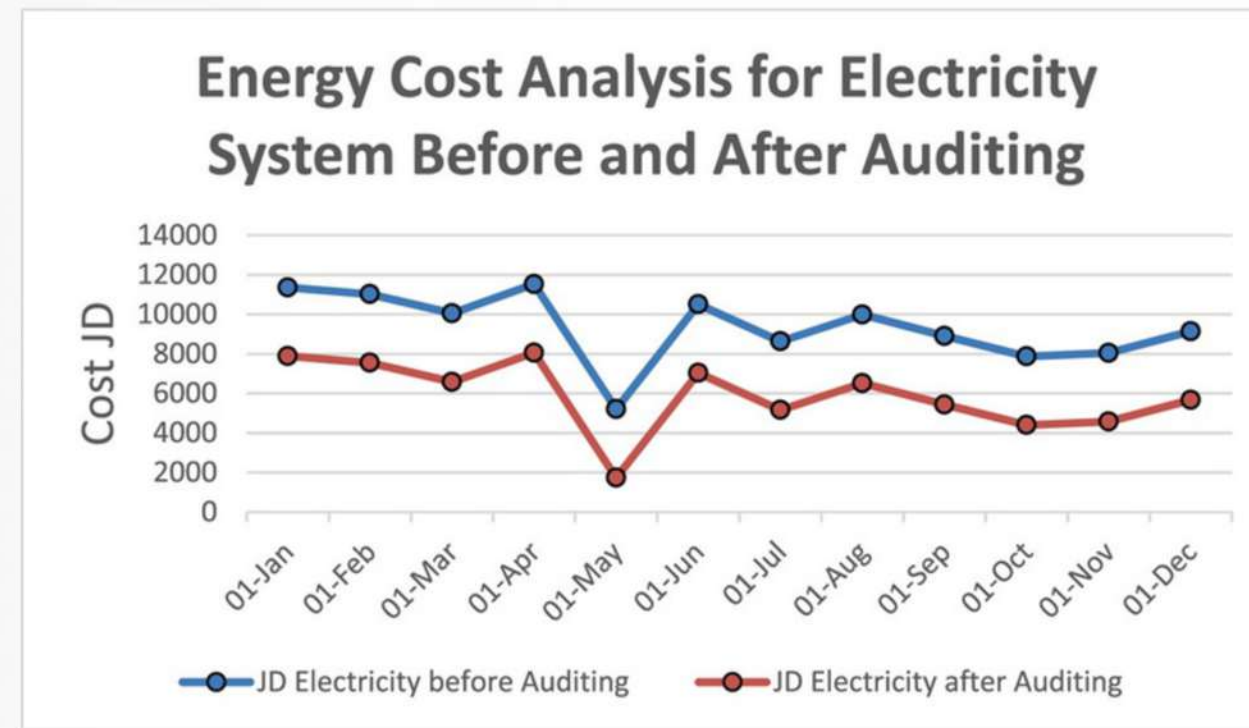
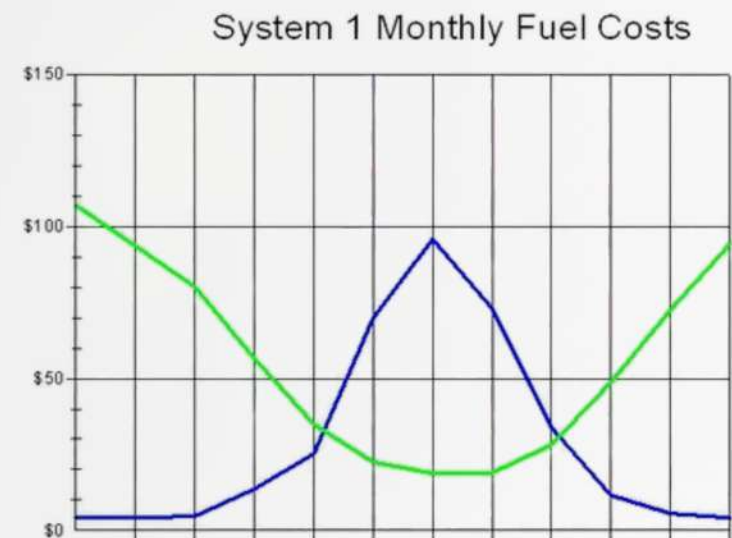
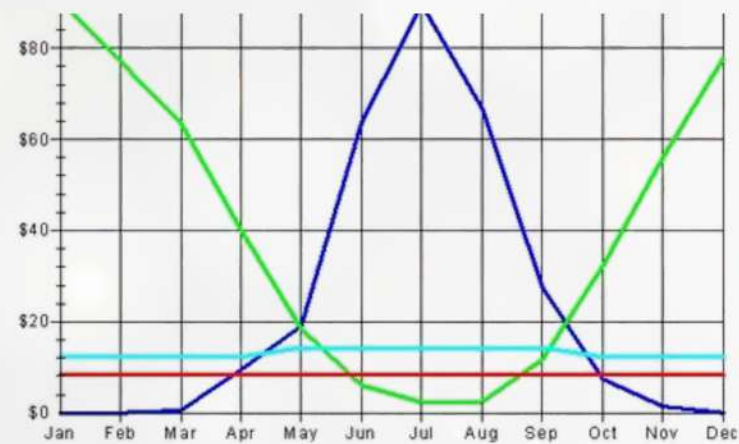


Cost-Saving Benefits by Sector



ENERVISION

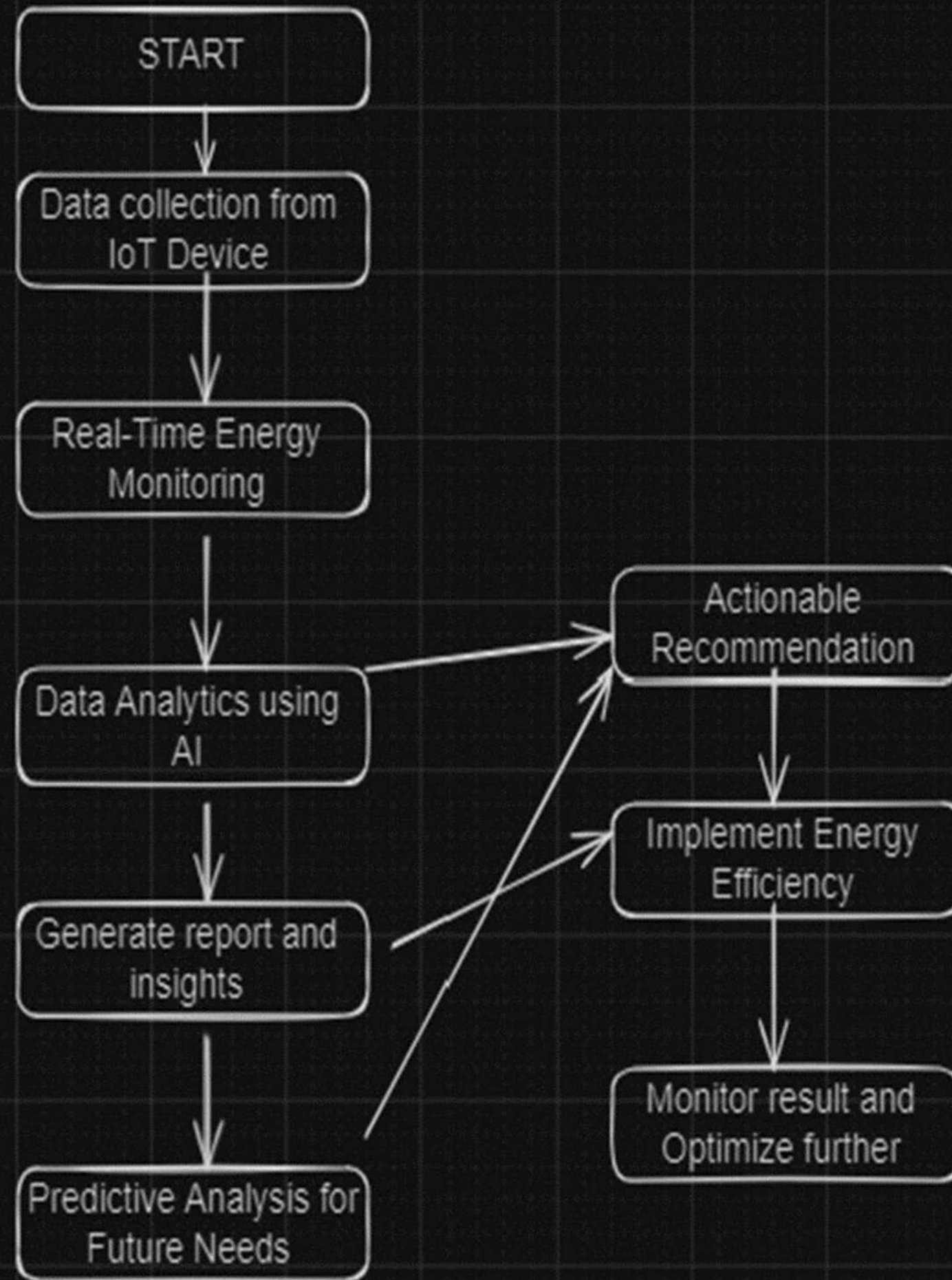
TECHNICAL STACK



- Leverages sensors to track energy consumption in **lighting, HVAC systems, and machinery**, allowing for the prompt detection and **rectification** of inefficiencies.
- Utilizes sophisticated machine learning algorithms to **identify usage patterns**, flag irregularities, and forecast **energy requirements** with a high degree of precision (**95% accuracy**).
- Ensures seamless data storage and processing on a centralized platform, offering scalable and **real-time access to energy analytics**, protecting through **encryptions**.
- User-friendly interface provides live updates, alerts, and actionable insights for efficient resource management.

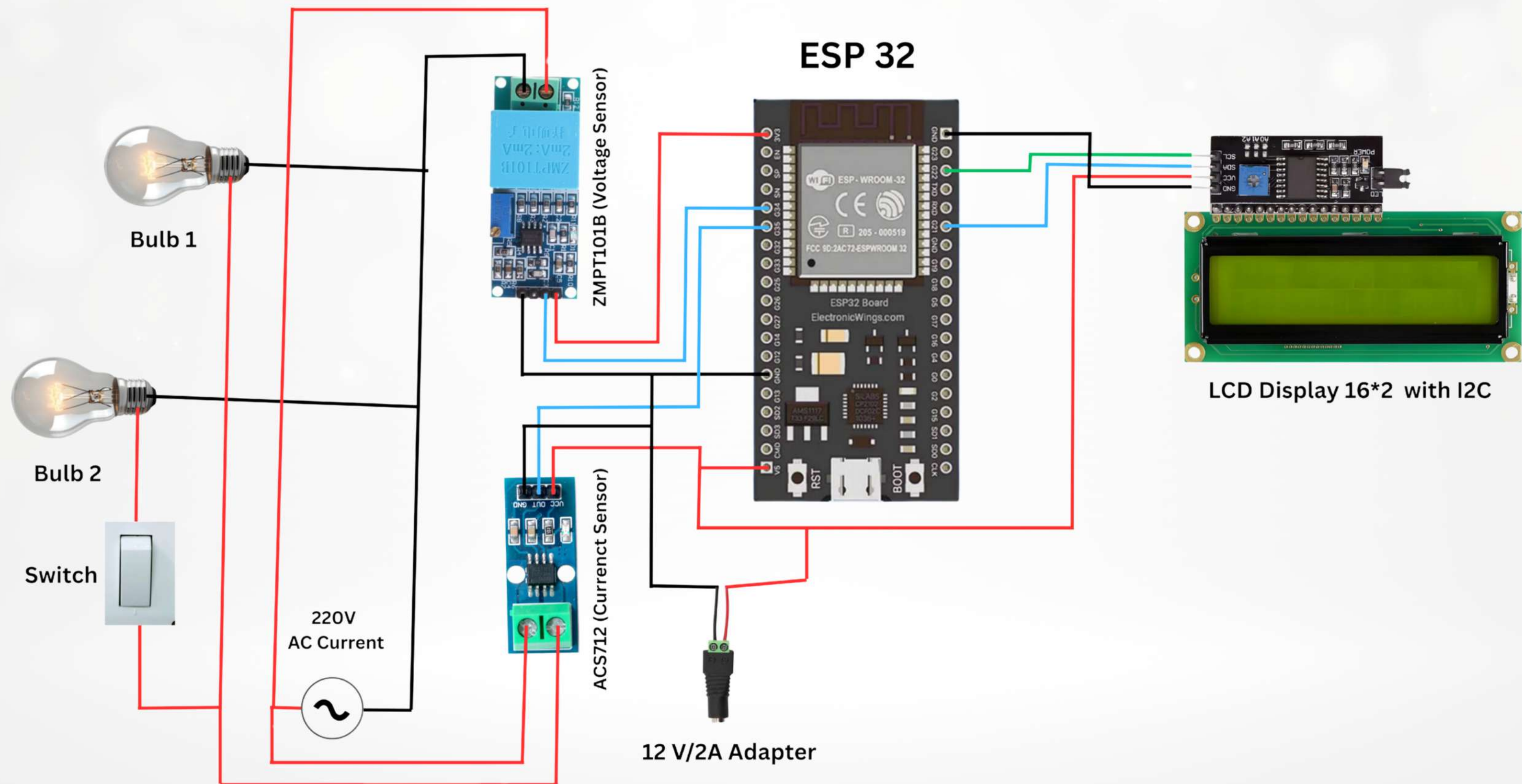


ENERVISION WORK FLOW TECHNOLOGY



- IoT sensors track real-time energy usage across all systems with encrypted data flow.
- AI detects inefficiencies, predicts demand with high accuracy, and flags anomalies instantly.
- Automated reports deliver insights, appliance-level breakdowns, and carbon credit generation data.
- Actionable recommendations optimize usage, reduce costs, and continuously improve performance.

ENERVISION IOT - CIRCUIT DIAGRAM



ENERVISION COMPETITION

- Global tools like Schneider EcoStruxure & Siemens Desigo focus on large enterprises, while EnerVision targets **mid-size to enterprise with faster deployment and lower setup cost.**
- Most competitors offer only monitoring dashboards, whereas EnerVision integrates **AI prediction, peak-hour intelligence, and appliance-level analytics in one unified platform.**
- Traditional energy audit firms perform manual, periodic audits; EnerVision provides real-time, automated, continuous auditing with AI-driven insights.
- Competitors lack architectural design analysis, while EnerVision identifies building-level structural inefficiencies (HVAC layout issues, load design, lighting zoning).
- Very few platforms support **carbon credit tracking & generation,** making EnerVision one of the only systems **offering energy optimization + new revenue stream creation.**

Feature/Aspect	EnerVision	Google Energy System	Indian Energy Audit System
Real-time IoT Monitoring	✓ Yes	✓ Yes	⚠ Partially
Predictive Analytics	✓ 95% accuracy	✓ Strong	⚠ Limited
Architectural Design Analysis	✓ Included	✗ No	✗ No
Carbon Credit System	✓ Full automation	✗ None	✗ None
Sector Customization	✓ High	✗ Low	⚠ Medium
Sustainability Reporting	✓ Advanced	✓ Strong	⚠ Inconsistent
Cost	₹8–15 Lakhs	₹20–40+ Lakhs	₹1–3 Lakhs
Energy Savings	20–25%	12–15%	8–12%
Ease of Use	Very Easy	Complex	Simple (but low insight)
Deployment Effort	Low	High	One-time

ENERVISION REVENUE STREAM

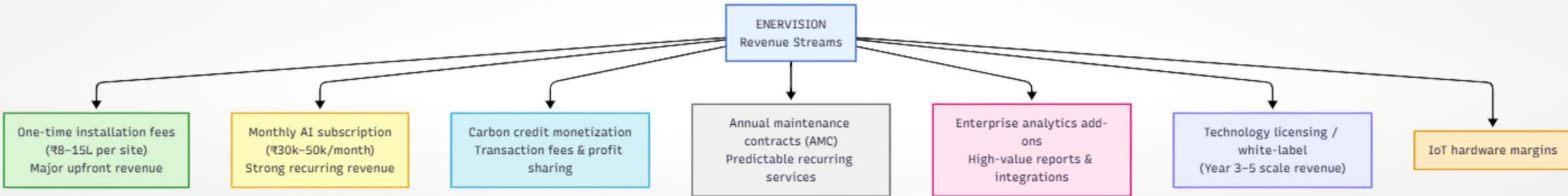
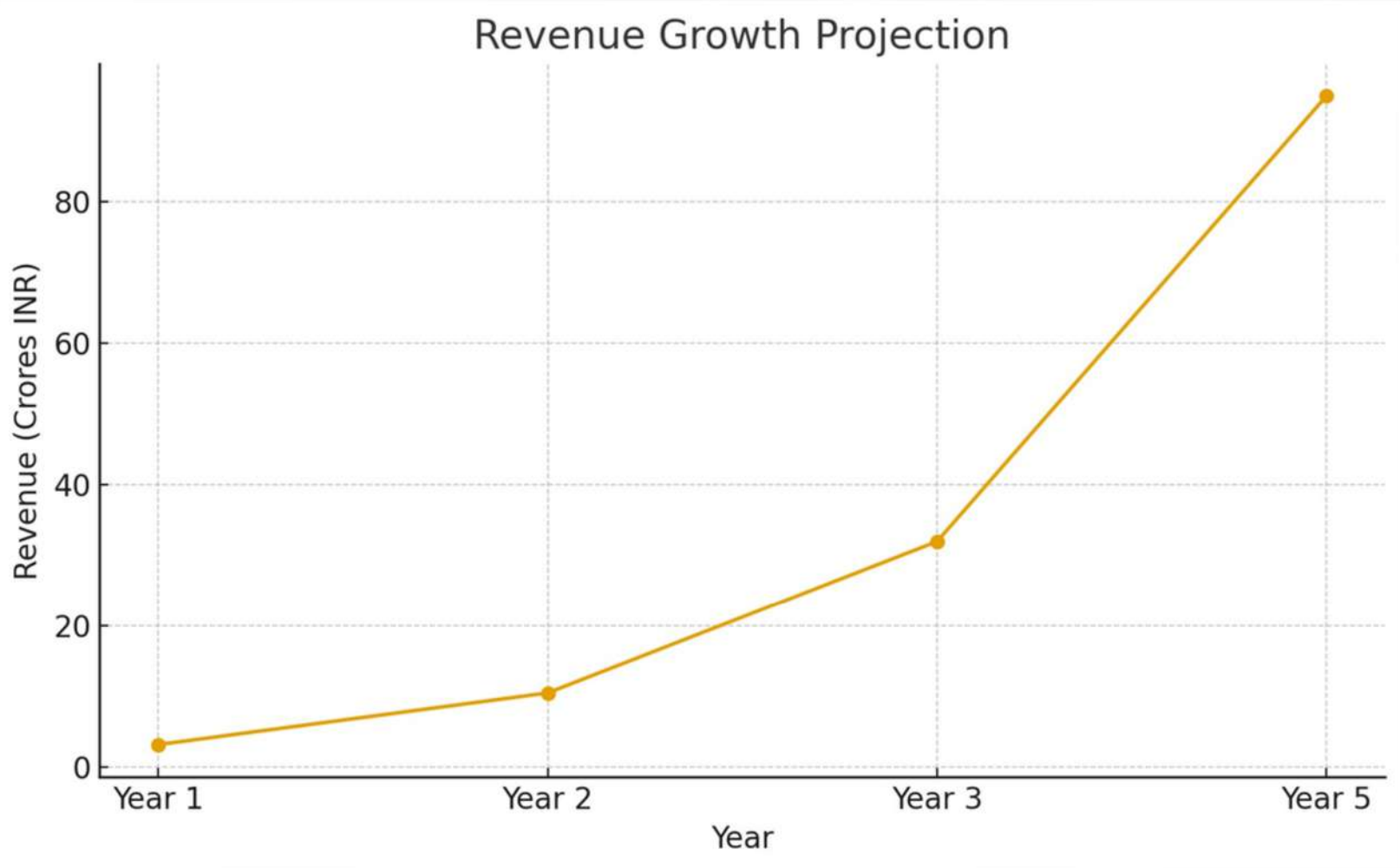
Year 1: 25 installations generating ₹3.2 crores revenue; EBITDA ₹40 lakhs (12.5%), proving business model viability and achieving cash-flow break-even by Q4.

Year 2: Scale to 75 installations and ₹10.5 crores revenue; EBITDA jumps to ₹3.3 crores (31%), demonstrating strong operating leverage and predictable subscription growth.

Year 3: 200+ installations delivering ₹32 crores revenue and ₹14 crores EBITDA (44%), supported by multi-city expansion and recurring subscription compounding.

Year 5: Over 600 installations with ₹95 crores revenue, recurring revenue reaching 40% (₹38 crores), and EBITDA stabilizing at 48–50%, indicating a highly scalable and profitable model.

Projections assume 60% retention, but **actual expected 85%+ retention would accelerate growth** further due to high switching costs and ongoing value delivery.



ENERVISION TRACTION

Verified Market Demand (Survey-Based): Over 80% of surveyed institutions and industries reported significant energy wastage, lack of real-time visibility, and high interest in continuous auditing solutions like EnerVision.

High Adoption Readiness: 3 out of 4 decision-makers (principals, facility managers, industrial supervisors) expressed willingness to implement EnerVision once full deployment is available—showing strong pre-market pull.

Prototype Validation: Initial prototype testing confirmed EnerVision’s ability to capture live energy data, flag anomalies, and generate actionable insights, validating technical feasibility and system reliability.

ROI-Driven Customer Motivation: Survey responses show customers are ready to adopt if they can achieve 15–25% yearly savings, with most expecting 6–12 months ROI, confirming strong economic motivation.

Carbon-Credit Advantage as a Differentiator : 70% of surveyed customers showed high interest in EnerVision’s carbon tracking and future carbon-credit generation, identifying it as a key value-add over traditional audits.



ENERVISION

RESOURCE MOBILIZATION

Category	Amount (₹)	% of Total	Description
Product Development & Prototyping	300,000	33%	IoT sensor enhancement, gateway development, architectural design analysis, hardware testing, and system reliability improvement.
Cloud, AI & Software Development	240,000	27%	Cloud hosting (AWS/Azure), API services, AI model training, forecasting engine, anomaly detection, dashboard UI/UX, and mobile integration.
Field Testing & Pilot Deployments	150,000	17%	Procurement of CT sensors, smart meters, gateways, travel for site visits, live environment testing in schools/offices/industries, user feedback collection.
Research, Survey & Carbon Credit Framework	100,000	11%	Market studies, stakeholder feedback, sustainability modelling, carbon footprint calculation, carbon-credit readiness architecture.
Outreach, Documentation & Certifications	110,000	12%	Technical documentation, proposal creation, exhibitions/ESG summits, BEE workshops, IP/certification filing.
Total	₹9,00,000	100%	—

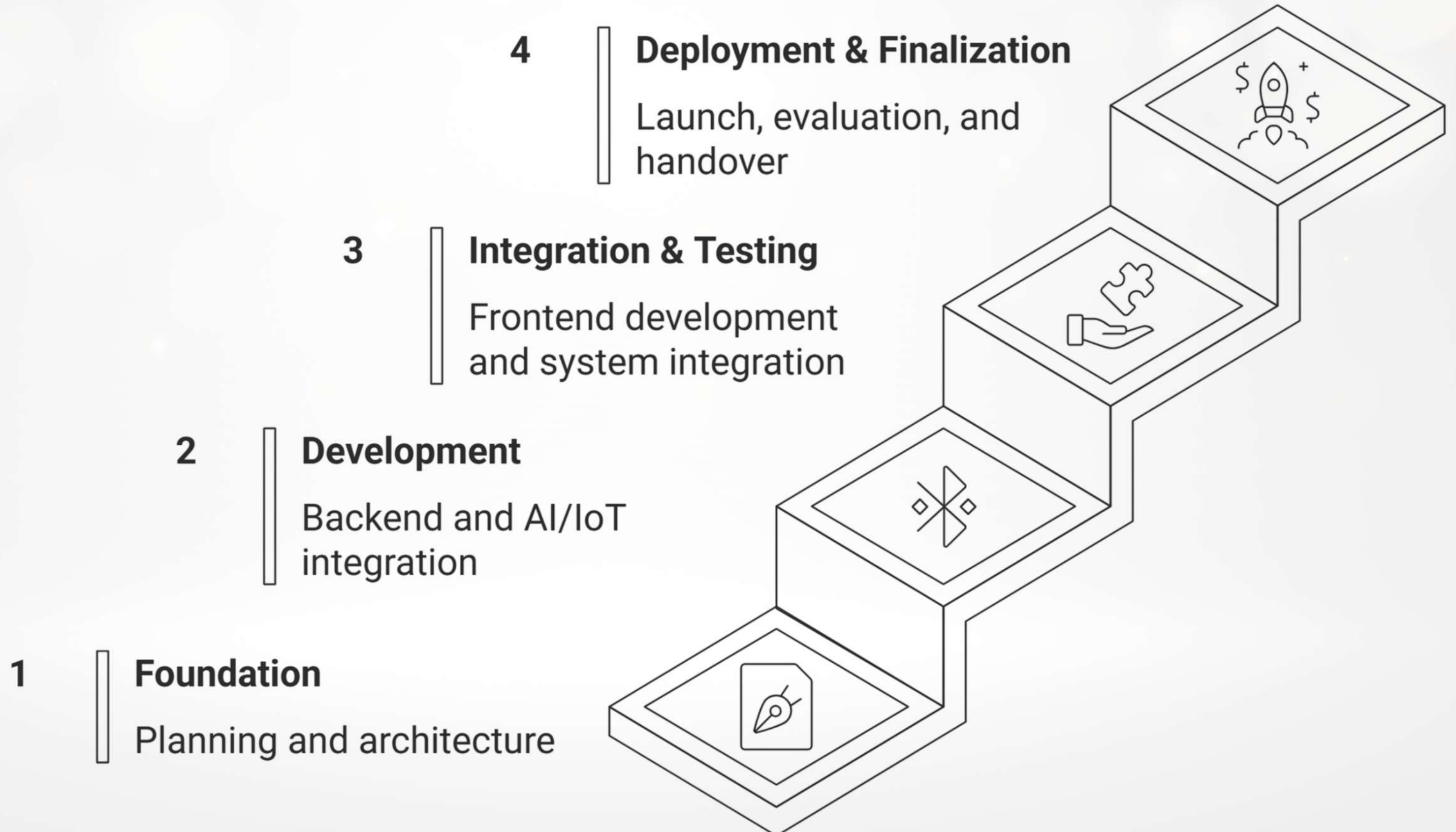


ENERVISION

RESOURCE MOBILIZATION

Human Resource	Fellowship-supported	Technical Development Effort
		Deployment & Coordination
		AI/ML & Carbon-Credit System Development
Technical Resources	Hardware procurement	IoT Sensors & Smart Meters
		Gateways & Communication Modules
	Setup investment	Energy Simulation Lab
	Operational cost	Cloud Infrastructure
Partnerships & Ecosystem Support	Collaboration	Educational Institutions
		MSME Manufacturing Units
	Expert partnership	Architects / Energy Design Experts
		Sustainability & Carbon Credit Consultants
		Energy Auditors / BEE Consultants
Expected Outcomes (2 Years)	Validated IoT + AI Prototype	
	Pilot Deployments	
	Carbon-Impact & Credit Framework	
	Market-Ready Solution	
	Customer Validation	

ENERVISION WAY FORWARD STRATEGY



MEET OUR **TEAM MEMBERS**



Bhuvanesh Ganapathi
Master of Computer Application
PSGCAS



Tharun Vaibhav S S
Master of Computer Application
PSGCAS



Prateepan MS
Bachelor of Computer Application
PSGCAS

A top-down view of a desk with various items: a laptop, a cup of coffee, a pen, a pencil, glasses, and a large green leaf. The text "THANK YOU" is centered in the middle of the image.

THANK YOU